

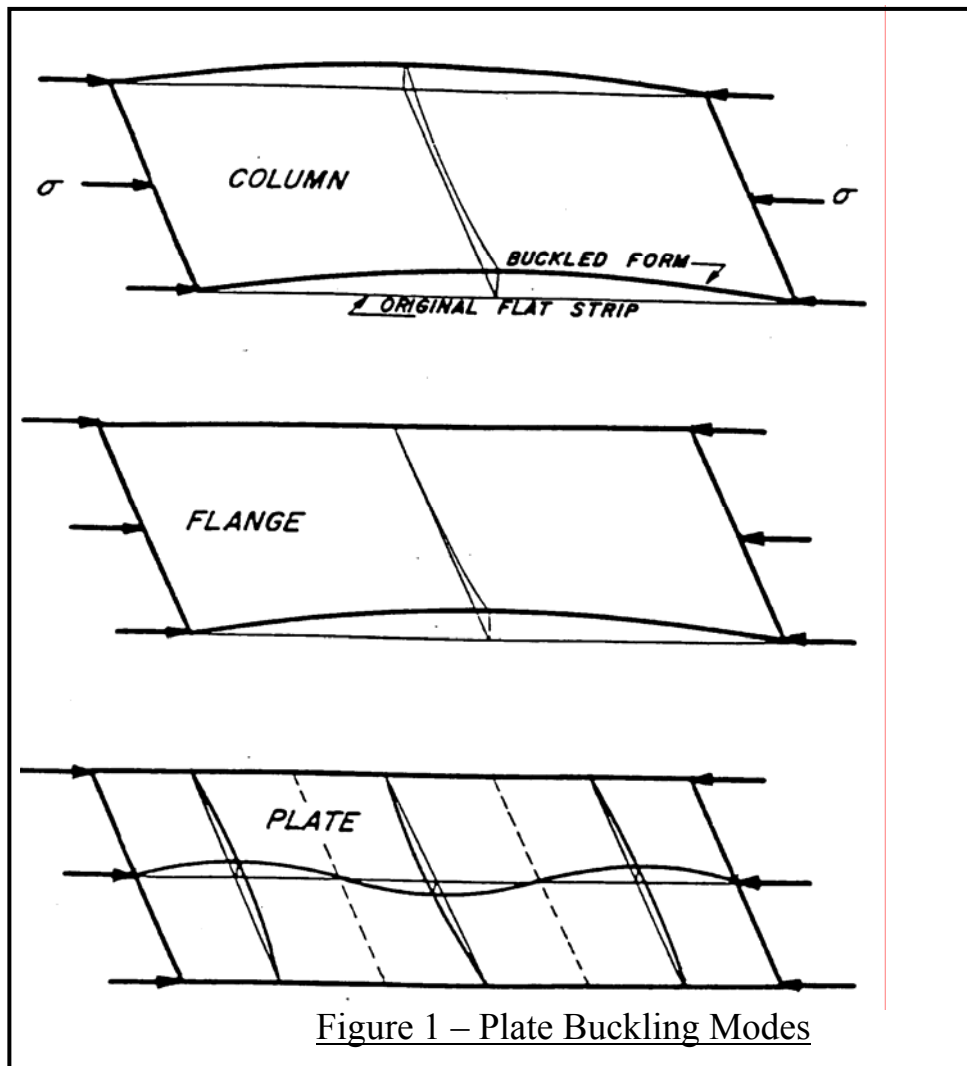
## COMPRESSION PANELS

### Introduction

This lecture addresses the basic theory of plate buckling, plasticity effects and the instability failure modes of skin/stiffener compression panels.

### Panel Buckling Modes

Figure 1 shows the buckling modes of a thin sheet when its edges are fixed in three different ways giving column, flange and plate configurations. For the column the sides are free and allowed to bow when the load is applied, and after the application of sufficient load the column will exhibit curvature in the longitudinal direction. If as in the plate the unloaded edges are supported, then curvature will also occur in the lateral direction. This curvature is known as Anticlastic Bending and adds to the rigidity of the plate.



### Simplified theory for initial buckling stress

The theoretical solution of the problem of instability of a flat plate is complex and therefore a simplified solution is used here to derive the expression for the buckling stress.

The flat plate ABCD in Figure 2 is loaded with a compressive load  $P$ , uniformly distributed over the edges AD and BC, so that the compressive stress is  $P/bt$ . Consider a narrow strip EF of width  $db$  and assume it to be given a small displacement such that the deflection is  $y$  at  $x$ .

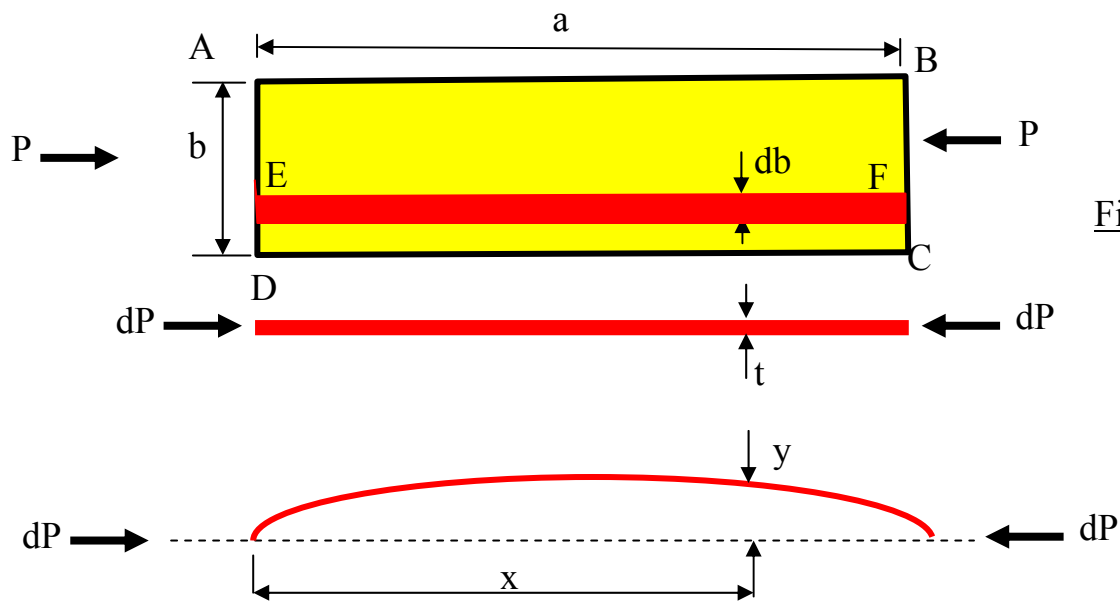


Figure 2

The compressive load on the strip is  $dP = \sigma_t db$  and the bending moment at  $x$  is given by  $M = dP \cdot y$

The longitudinal stress due to bending is  $\sigma_x = (Mt/2)/(dI)$ , where  $dI$  is the second moment of area of the cross section of the strip.

If the strip was completely isolated, the longitudinal strain  $\sigma_x/E$  would be accompanied by a lateral strain  $-\nu\sigma_x/E$ . Due to the proximity of the other strips, however, the lateral strain is prevented. (This effect will always arise if the width of the plate is large compared with its thickness). The prevention of this strain is equivalent to subjecting the strip to a lateral strain of  $+\nu\sigma_x/E$ , which in turn produces a further longitudinal strain of amount  $-\nu^2\sigma_x/E$ .

Hence the total longitudinal strain  $\epsilon_x = \frac{\sigma_x}{E} - \nu^2 \frac{\sigma_x}{E} = \frac{\sigma_x}{E}(1 - \nu^2) \dots \text{equ.1}$

Substituting the value of  $\sigma_x$  from above then  $\epsilon_x = \frac{Mt}{2E} \left( \frac{1 - \nu^2}{dI} \right) \dots \text{equ.2}$

If R is the radius of curvature of the neutral surface, assumed to be the mid-surface of the plate, it can be shown that providing plane cross sections remain plane then:

$$\epsilon_x = \frac{t/2}{R} \dots \text{equ.3} \quad \text{also} \quad \frac{d^2y}{dx^2} = \frac{1}{R} = \frac{\epsilon_x}{t/2} \dots \text{equ.4}$$

From equations 1, 2, 3 and 4 substituting for R :-

$$\frac{d^2y}{dx^2} = \frac{M(1 - \nu^2)}{EdI} \quad \text{writing} \quad E' = \frac{E}{1 - \nu^2} \quad \text{and since } M = ydP \quad \text{then}$$

$$\frac{d^2y}{dx^2} = \frac{ydP}{E'dI} \dots \text{equ.5}$$

Equation 5 is in the same form as the Euler equation for struts hence the critical value of dP is:-

$dP_{\text{CRIT}} = \frac{\pi^2 E' dI}{a^2}$  and since  $dI = \frac{db \times t^3}{12}$  and  $dA = t.db$  then the critical buckling stress is:-

$$\sigma_{\text{CRIT}} = \frac{dP_{\text{CRIT}}}{dA} = \frac{\pi^2 E}{12(1 - \nu^2)} \left( \frac{t}{a} \right)^2 \dots \text{equ.6}$$

This expression holds for all the strips and can be considered to be valid for the whole plate.

The conditions for Equation 6 are for when the unloaded edges (sides) are free and the loaded edges (ends) are simply supported and free to rotate but not free to deflect normal to the plate.

For a plate simply supported on all four edges the buckling compression load is considerably higher, as the plate deflects both longitudinally and laterally with the lateral strips having to bend. The supporting effect of the lateral strip may be sufficient to cause a lengthwise strip to deflect into two or more waves as shown in Figure 1.

The simple analysis considered does not account for the deflection  $y$  depending on  $z$  the distance from edge CD, as well as on  $x$ . In the detailed theory (Ref Timoshenko: Theory of Elastic Stability) a deflection curve of the form:-

$$y = y_{\max} \sin\left(\frac{nx\pi}{b}\right) \sin\left(\frac{mz\pi}{a}\right) \text{ is assumed and it can be shown that}$$

$$\sigma_{\text{CRIT}} = \left( \frac{a}{mb} + \frac{mb}{a} \right)^2 \frac{\pi^2 E}{12(1-\nu^2)} \left( \frac{t}{b} \right)^2 \dots \text{equ.7}$$

where  $m$  = the number of half waves in the  $x$  (length) direction

Equation 7 may be written as  $\sigma_{\text{CRIT}} = KE \left( \frac{t}{b} \right)^2$  where  $K$  is the buckling stress

coefficient and is a function of  $a/b$  (ratio of the sides), the type of side and end fixings and  $\nu$ .

The value of  $K$  can be obtained from standard data sheets, such as the one given in Figure 4. Note: Beyond an aspect ratio around 3.0, the effect of the end fixings is dominated by that of the sides, with the values of  $K$  converging to a single value commensurate with the side fixation (one free, simple supports, fixed).

In the practical world, it is difficult to obtain edges that are either simple supports or fixed, with the practical solution being somewhere in between. A judgement has to be made of the degree of fixation offered by the supporting member. Figure 3 illustrates this showing two different types of stringers/stiffeners and hence edge fixings.



Figure 3

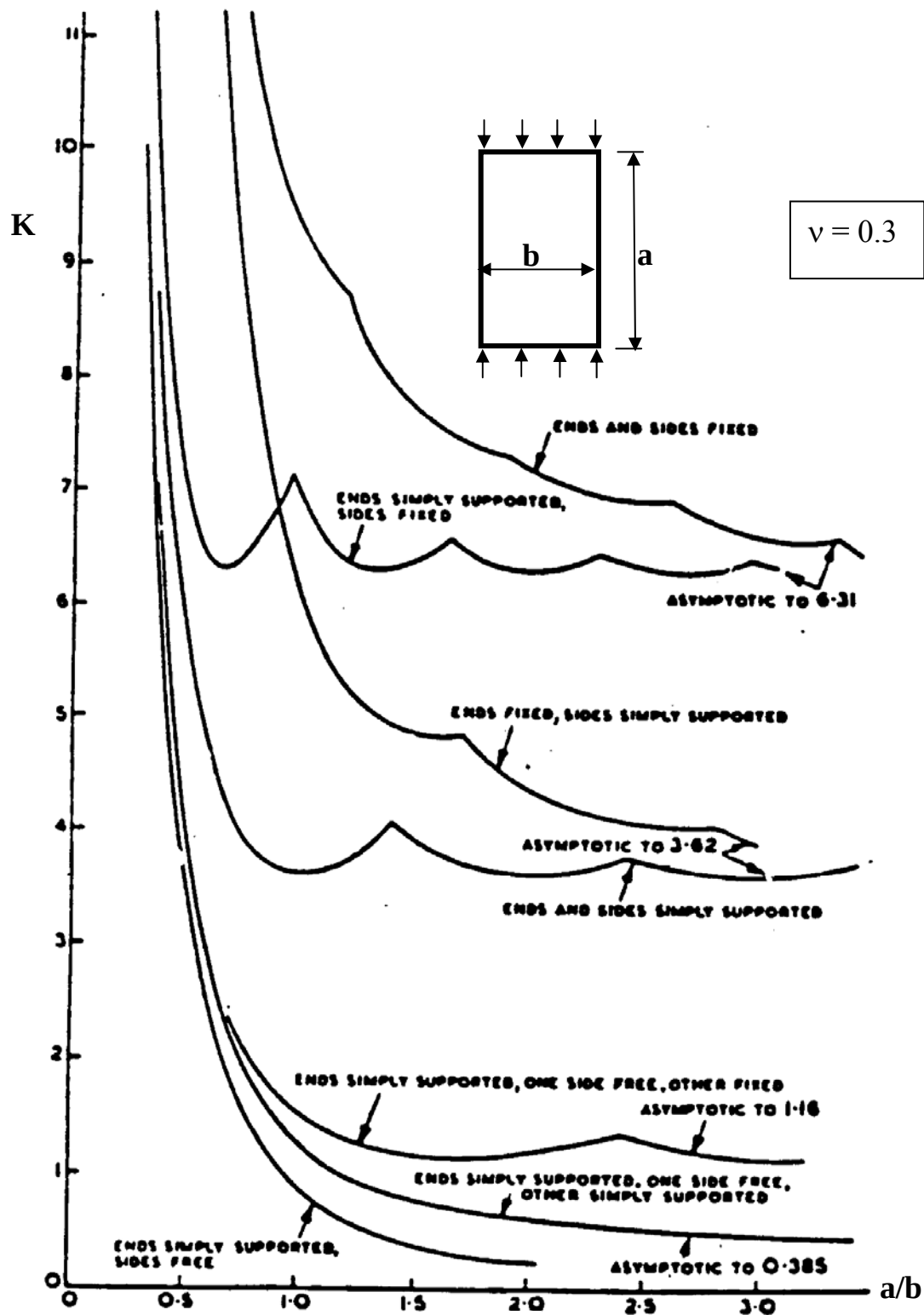


Figure 4 – Buckling stress coefficients for flat plates in compression

## See Tutorial Question 1 (part (a) only)

### Tangent Modulus method of accounting for material plasticity effects

In light alloys the yield point is not as clear cut as with steels and there is a gradual decrease in the value of 'E' at the higher stress levels. We will now examine the effect of this change on strut and plate buckling values and see how it is dealt with using Euler strut buckling theory. Figure 5 shows the Euler hyperbola and the actual failure curve. Figure 6 shows a typical stress – strain plot for a light alloy.

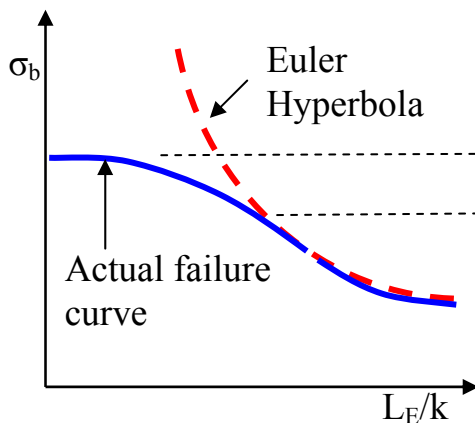


Figure 5

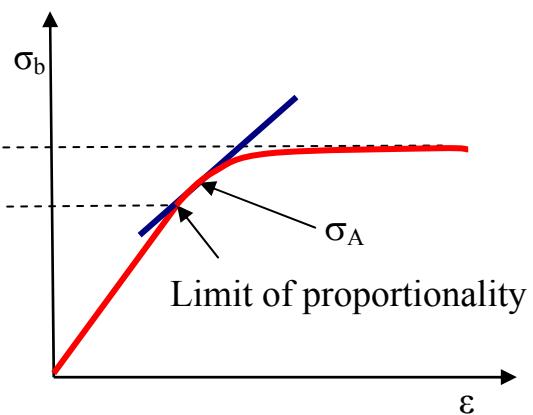


Figure 6

The Figures show that upto the limit of proportionality for the material the Euler formula gives an accurate prediction for the buckling stress but as the E value starts to decrease at a higher stress level then the Euler formula no longer holds. The tangent modulus ( $E_T$ ) of the material is defined by the slope of the tangent to the stress –strain plot, hence the the tangent modulus at  $\sigma_A$  is the slope of the tangent line shown in Figure 6.

## See Tutorial Question 2

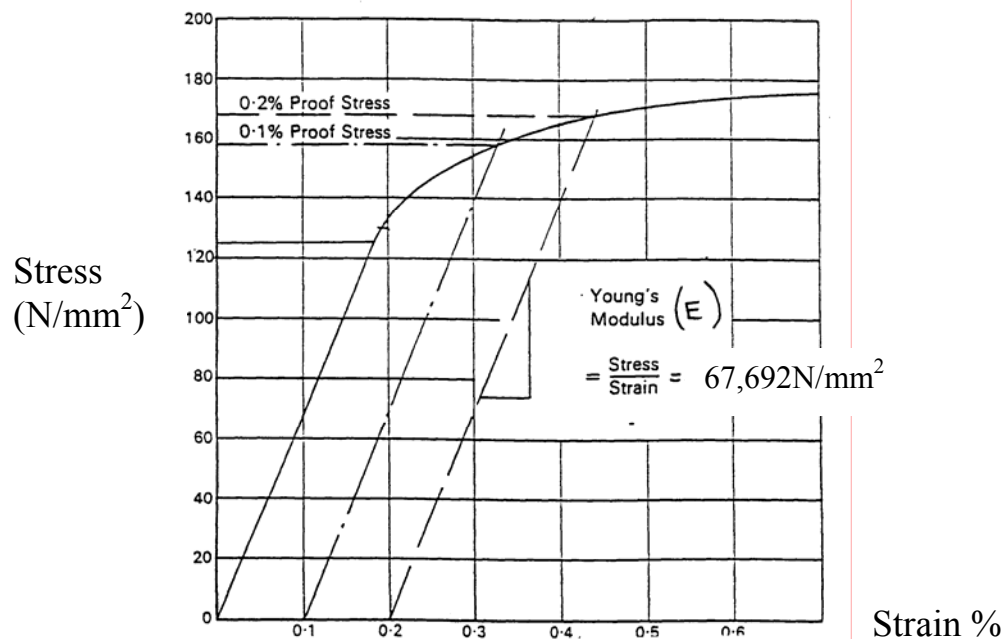


Figure 7 – Plot of Stress v % Strain for an aluminium alloy

The predicted buckling stress  $\sigma_e$  using Young's Modulus and the buckling stress using the tangent modulus at  $\sigma_e$  are incompatible.

Clearly the true buckling stress will lie between about 125 N/mm<sup>2</sup> and 160 N/mm<sup>2</sup> and this value will be where the predicted buckling stress for a certain value of  $E_T$  will be the same as the stress at which this  $E_T$  occurs

This buckling stress could be determined by a series of iterations as illustrated by the following considerations :-

Now a stress of 159 N/mm<sup>2</sup> will cause buckling failure if  $E = 67,692 \text{ N/mm}^2$  so

- Load (stress) strut up to 60 N/mm<sup>2</sup> and it will not buckle as  $E = 67,692$
- Load (stress) strut up to 125 N/mm<sup>2</sup> and it will not buckle as  $E = 67,692$
- Load strut up to 126 N/mm<sup>2</sup> and it will not buckle but  $E < 67,692$  so the buckling stress is now less than 159 N/mm<sup>2</sup>, say 150.
- Load strut up to 127 N/mm<sup>2</sup> it still will not buckle but  $E$  has reduced even lower so buckling stress is now less than 150 N/mm<sup>2</sup>, say 140.
- Increase load on strut up to 128 N/mm<sup>2</sup>, it will now just buckle as  $E = 54,189 \text{ N/mm}^2$  which when used in the EULER buckling equation gives 128 N/mm<sup>2</sup> hence the values are now compatible.

This stress can, however, be more easily determined if we have a plot of Stress v (Stress/  $E_T$ ). Taking the material Stress v Strain plot shown in Figure 7, then considering the  $E$  and  $E_T$  values at stresses 60, 125, 130, 140, 150, and 160  $\text{N/mm}^2$ , we can compile table 1:-

| $\sigma (\text{N/mm}^2)$ | $E \text{ or } E_T (\text{N/mm}^2)$ | $\sigma / E_T$ |
|--------------------------|-------------------------------------|----------------|
| 60                       | 67,692                              | 8.8E-4         |
| 125                      | 67,692                              | 18.45E-4       |
| 130                      | 53,379                              | 24.4E-4        |
| 140                      | 29,955                              | 46E-4          |
| 150                      | 17,465                              | 85.9E-4        |
| 160                      | 11,866                              | 134.8E-4       |

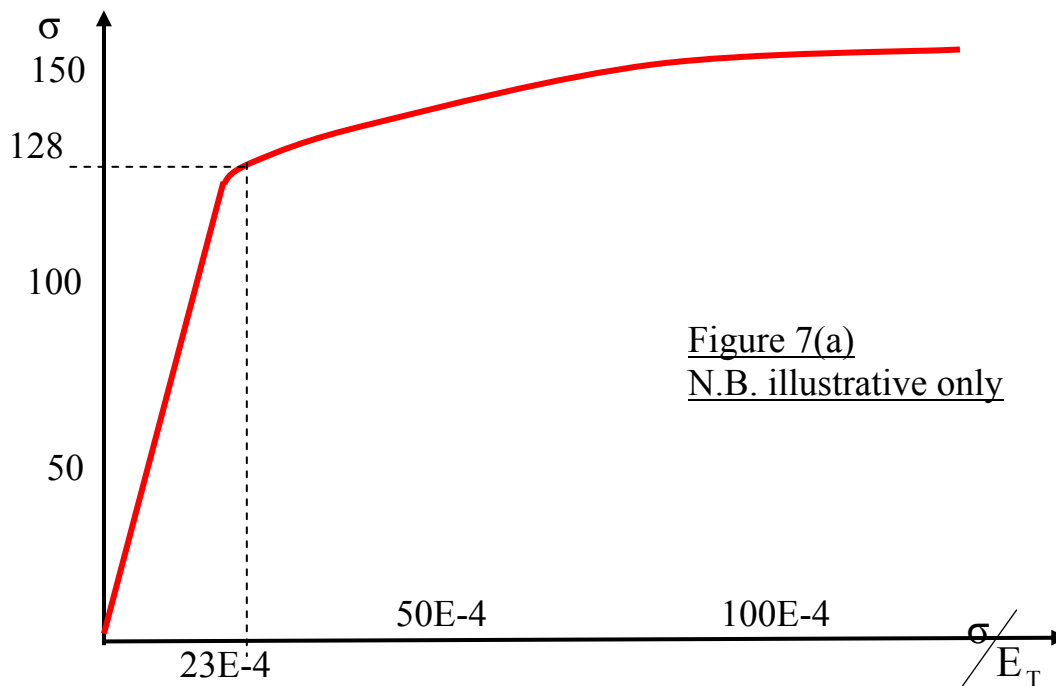
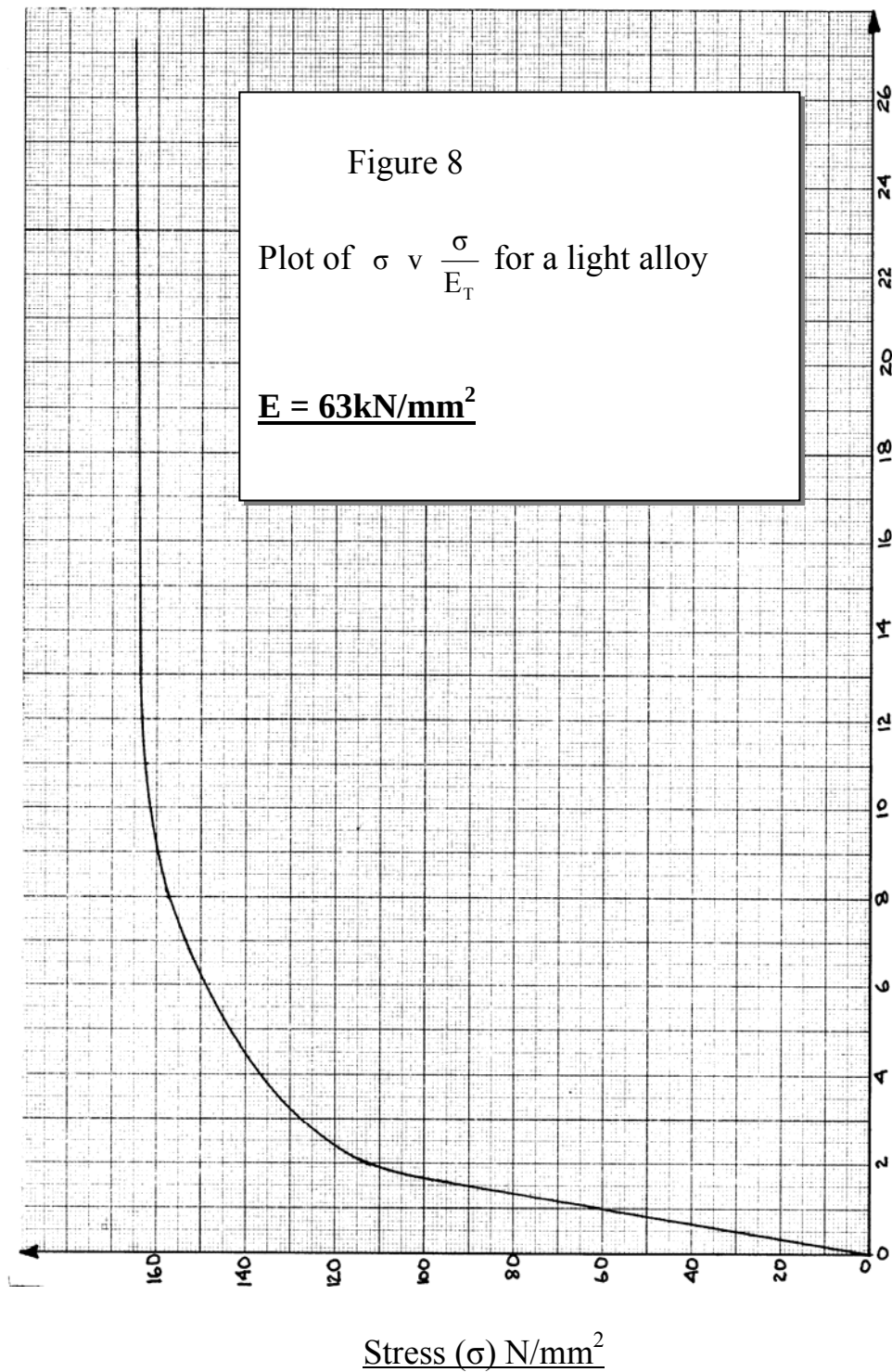


Figure 7(a)  
N.B. illustrative only

By plotting  $\sigma$  against  $\sigma / E_T$ , Figure 7(a) is obtained. Hence by calculating the value of the ratio  $\sigma_b / E_T \left( = \frac{n^2 \pi^2 I}{AL^2} \right)$  the buckling stress can be read directly from Figure 7(a). A plot for a light alloy is shown in Figure 8 for later use.

**See Tutorial Question 1 (part (b))**



## Panel Skin Buckling

Up to the initial buckling stress it can be assumed that the stress is constant (uniform) in the skin and the stringer, given that they are of the same material.

After the skin buckles any increase in load would increase the stress at the outer edges of the panel, at the stringers, with the stress at the centre of the panel staying approximately constant at  $\sigma_b$

In the limit the edge stress, designated  $\sigma_e$  would reach a critical value at which the panel will collapse via a particular mode of instability. The stress in the skin in the post buckled state will not be uniform as indicated in Figure 9.

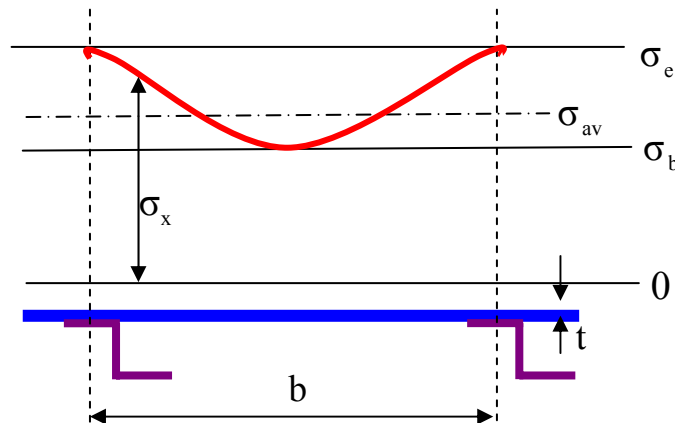


Figure 9

$$\text{Total load carried by the skin} = \Sigma \sigma_x (\delta b t) = \sigma_{av} b t$$

The total load capacity of one skin /stiffener combination (shown in Figure 10) may now be determined.

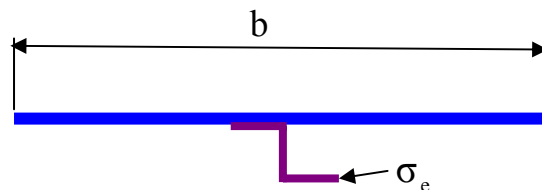


Figure 10

$$\text{Total load carried by one stringer and one pitch of skin} = \sigma_{av} b t + \sigma_e A_s$$

It has been found from tests that there is a fairly linear relationship between  $\sigma_{av}$  and  $\sigma_e$  remembering that up to buckling  $\sigma_{av}$  equals  $\sigma_e$  ie:-

$$\frac{d\sigma_{av}}{d\sigma_e} = \text{constant (0.33 for a parabolic stress distribution)}$$

between  $\sigma_e$  and  $\sigma_b$  which is often assumed.)

hence 
$$\sigma_{av} = \sigma_b + \frac{d\sigma_{av}}{d\sigma_e}(\sigma_e - \sigma_b)$$

Given that the maximum edge stress can be determined, then the maximum load for the skin stringer combination and the load carrying capacity of the skin per mm width can be calculated.

Load for one skin-stringer combination = 
$$\sigma_e A_s + [\sigma_b + \frac{d\sigma_{av}}{d\sigma_e}(\sigma_e - \sigma_b)]bt$$

The task remaining is to determine the value of  $\sigma_e$ , this will be given by the lowest of the allowable stresses calculated, there being a number of possible failure modes:-

- Local instability
- Inter rivet buckling
- Flexural instability

Each of these modes will be considered in more detail. However, there are further modes of failure, which may have to be considered, which briefly are:-

**Wrinkling** – (forced crippling) this is similar to inter-rivet buckling, but the skin and attached flange of the stringer buckle together in a mode longer than the fastener pitch.

**Torsional instability** – this can be an important mode for open sections, Z, C or L, the stringers twist over their effective pin length. For symmetric and closed section this mode can normally be ignored.

### Local Instability

This is one of the most common modes of failure in short panels or struts. Local buckling of either of free flange, web or attached flange of the stringer occurs as shown in Figure 11.

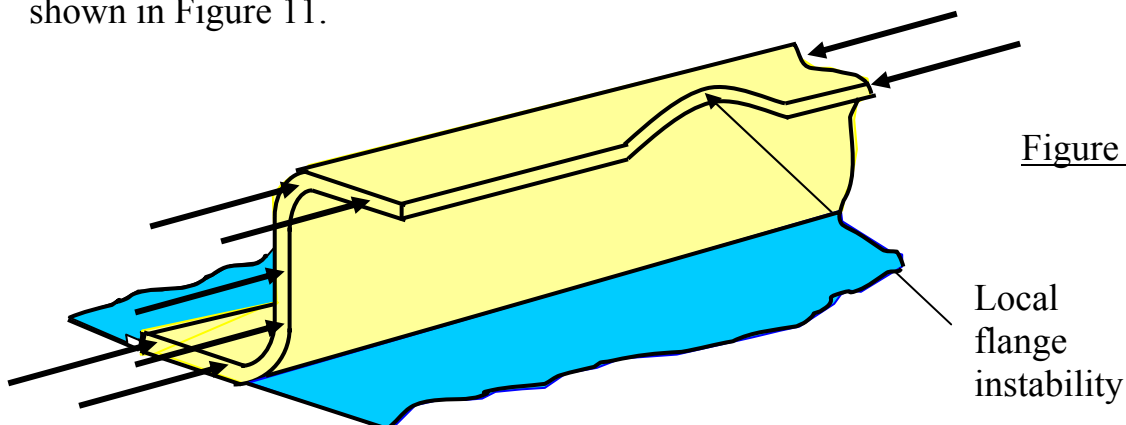


Figure 11

Buckling coefficients of 0.58 for the free flange and 3.62 for the web are taken for I, Z and C sections. The coefficient of 0.58 is used, as the support at the flange to web is considered better than a pure simple support, but not as good as a fixed edge. The use of bulbed or lipped sections, in order to stabilise the free flange is often found.

The local instability stress is calculated from the equation:-  $\sigma_{\text{CRIT}} = KE \left( \frac{t}{b} \right)^2$

Values of K for different types of stiffener can be obtained from ESDU sheets.

### Inter-Rivet Buckling

This is an Euler type failure which can occur in the skin if the rivet pitch is sufficiently large, the skin buckling as a strut, hence:-

$$\sigma_{\text{IR}} = \frac{n^2 \pi^2 E b t^3}{L^2 \times 12 b t} = \frac{K_r \pi^2 E t^2}{12 p^2} \quad \text{where } p = \text{rivet pitch}$$

The rivet head produces a degree of fixation,  $K_r$  for a number of different rivet attachments the value of  $K_r$  is given in table 1 below.

| Type of attachment     | Fixity coefficient $K_r$ |
|------------------------|--------------------------|
| Flat head              | 4                        |
| Mush or snap           | 3                        |
| Countersunk or dimpled | 1.5                      |

Table 1—Rivet fixity coefficients

The value of  $K_r$  comes from the Euler fixity coefficient, e.g. for fixed ends  $K_r = 4$ .

The rivets attaching the skin to the stringers are performing two functions namely

- the rivets allow for the load to be passed in to and out of the stringer; in acting as a shear connection the shear forces are normally small.
- the rivets constrain the skin and stringer to act as one structural member, the skin-stringer, by preventing the skin from buckling off the stringer.

Inter rivet buckling is most likely to occur if the skin is thin in comparison to the stringer.

## Flexural Instability

This again is an Euler type of failure, involving the overall instability of the panel. To determine the flexural instability allowable, the stringer and an associated amount of 'effective' skin will have to be considered as a skin-stringer combination.

In order to calculate the flexural instability stress we must consider the stringer as a strut having an 'effective' width of skin  $w$ , working with it. Consider that there is a width of skin either side of the stringer working with it as a strut. There will remain a portion of skin in the centre of the panel that is not effective, or apparently absent as shown in Figure 12.

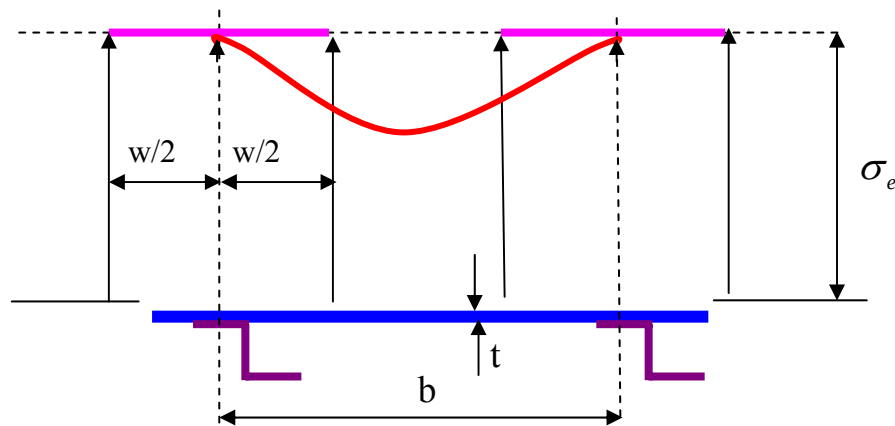


Figure 12

For design purposes an initial value of  $30t$ , where  $t$  is the skin thickness, is often assumed.

$$\sigma_{av} bt = \text{load in in the skin} = \sigma_e wt$$

$$\text{hence } w = \frac{\sigma_{av}}{\sigma_e} b$$

This width of skin is then assumed to act with the stringer and the second moment of area of the combination is then used in determining the flexural failure stress.

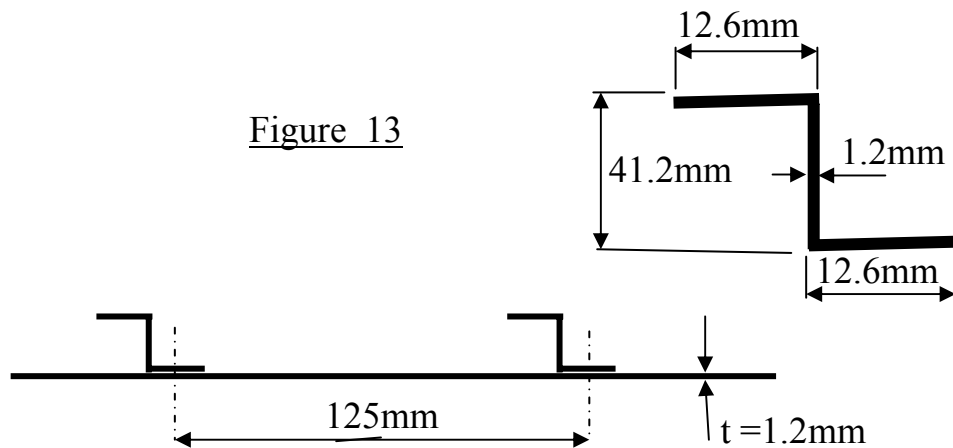
$$\sigma_e = \frac{n^2 \pi^2 EI}{AL^2} = \frac{n^2 \pi^2 E A k^2}{AL^2} = \frac{\pi^2 E}{\left(\frac{L}{k}\right)^2} \quad \text{for pin jointed ends.}$$

### Example

For the panel shown in Figure 13 determine;

- the critical stress values
- the effective skin width working with the stiffener
- the total load capacity of the section.

Rib pitch is 800mm. For an effective skin width of  $30t$ ,  $k$  is 15.5mm.  $\frac{d\sigma_{av}}{d\sigma_e} = .28$



$$K(\text{skin}) = 4.1, \quad k(\text{web}) = 3.62, \quad k(\text{free flange}) = 0.58$$

For the free flange:

$$\frac{\sigma_{LI}}{E_t} = 0.58 \left( \frac{1.2}{12} \right)^2 = 5.8 \times 10^{-3}$$

$$\sigma_{LI} = 148 \text{ N/mm}^2$$

For the web:

$$\frac{\sigma_{LI}}{E_t} = 3.62 \left( \frac{1.2}{40} \right)^2 = 3.25 \times 10^{-3}$$

$$\sigma_{LI} = 129 \text{ N/mm}^2$$

Flexural instability of skin/stringer combination for  $w = 30t$

$$\frac{\sigma_{FI}}{E_t} = \frac{\pi^2}{\left(\frac{L}{k}\right)^2} = \frac{\pi^2}{\left(\frac{800}{15.5}\right)^2} = 3.7 \times 10^{-3}$$

$$\sigma_{FI} = 132 \text{ N/mm}^2$$

Skin buckling:

$$\sigma_{bSKIN} = 4.1 \times \left(\frac{1.2}{125}\right)^2 \times 63 \times 10^3$$

$$\sigma_{bSKIN} = 23.8 \text{ N/mm}^2$$

now

$$\sigma_{av} = \sigma_b + \frac{d\sigma_{av}}{d\sigma_e}(\sigma_e - \sigma_b)$$

$$= 23.8 + 0.28(129 - 23.8)$$

$$= \underline{53.3 \text{ N/mm}^2}$$

$$w = \frac{\sigma_{av}}{\sigma_e} \times b = \underline{51.6 \text{ mm}} \quad 51.6/1.2 = 43t \geq 30t \quad \therefore \text{O.K.}$$

Load capacity of one skin/stringer panel:-

$$P = \sigma_{av}bt + \sigma_{eMIN}A_s = 53.3 \times 125 \times 1.2 + 129 \times 76.8 [\text{ie } 12.6 \times 2 \times 1.2 + 38.8 \times 1.2]$$

$$= 7995 + 9907$$

$$= \underline{17902 \text{ N (17.9 kN)}}$$

NOTE: The minimum buckling stress may be higher than the proof stress, the proof stress then becoming the limiting factor.

## Wing ultimate load case

A wing structural cross section is shown in figure 14. The corner members are considerably larger in area than the stringers. With increasing bending load, the skin and stringers in compression will eventually buckle, but the thicker corner members will still be below their failure loads and so the load can be further increased until these members fail causing failure of the wing. In the state where the skin and stringers have buckled then Hooke's law no longer applies to the section and so a modified cross-section is used to determine the stresses known as the **effective cross-section**.

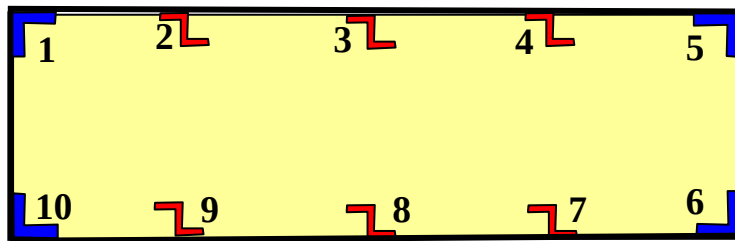


Figure 14

Consider the case of a beam, shown in figure 14(a), composed of two different materials (denoted by subscripts 1 and 2) which are connected rigidly together, then the strain will be proportional to the distance from the common neutral axis.

$$\text{Thus strain } (\epsilon) = \frac{\sigma_1}{E_1} = \frac{\sigma_2}{E_2} \quad \text{at any common interface} \dots \dots \dots (1)$$

Also, in general, as stress =  $E \times$  strain and strain is proportional to distance from the neutral axis in both materials, then:-

$$\sigma'_1 = E_1 y_1 K \quad \text{and} \quad \sigma'_2 = E_2 y_2 K$$

where  $\sigma'_1$  is the stress in material 1 at a distance  $y_1$  from the neutral axis etc.

$$\text{Hence} \quad \frac{\sigma'_1}{\sigma'_2} = \frac{E_1 y_1}{E_2 y_2} \dots \dots \dots (2)$$

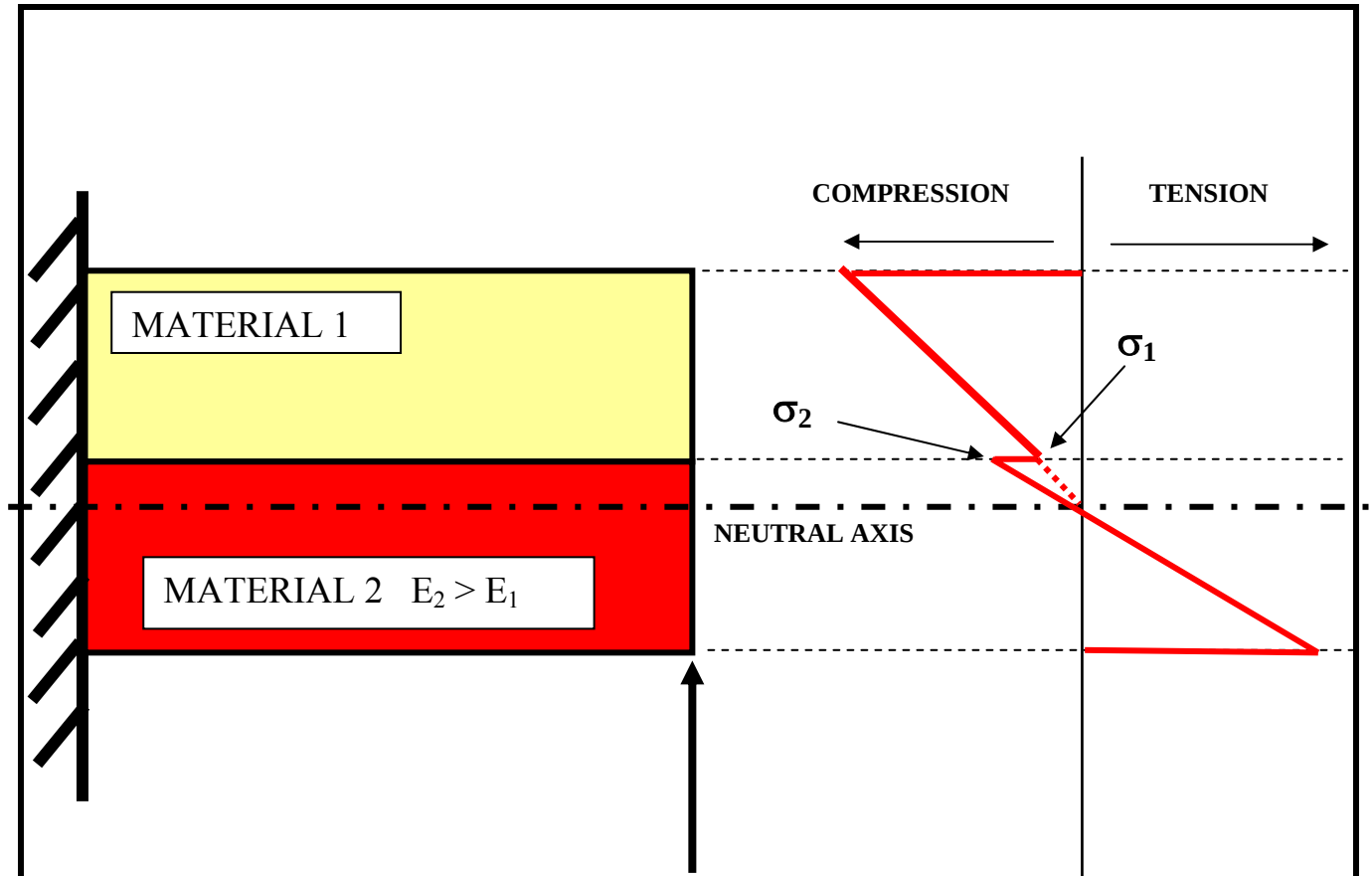


Figure 14a

Now  $\frac{M}{I} = \frac{\sigma}{y}$  thus  $M = \left(\frac{\sigma}{y}\right)I \propto I$

Therefore the amount of bending moment carried by each material is proportional to the  $I$  value of the material about the neutral axis.

Hence if  $I_1$  is the second moment of area of material 1 about the neutral axis and  $I_2$  is that for material 2 then:-

$$M_1 = \frac{\sigma_1 I_1}{y} \text{ (moment carried by material 1)} \quad \text{and} \quad M_2 = \frac{\sigma_2 I_2}{y}$$

where  $y$  is the distance from the neutral axis to the common surface,  $\sigma_1$  and  $\sigma_2$  are as equation 1 and  $M_1$  and  $M_2$  are the moments carried by materials 1 and 2 respectively.

The total moment of resistance is therefore:-

$$\begin{aligned} M &= M_1 + M_2 = (\sigma_1 I_1 + \sigma_2 I_2)/y \\ &= \left( \frac{\sigma_1}{y} \right) \left[ I_1 + \left( \frac{E_2}{E_1} \right) I_2 \right] \\ &= \left( \frac{\sigma_1}{y} \right) [I_1 + m I_2] \end{aligned}$$

$$\text{Therefore } \sigma_1 = \frac{My}{[I_1 + m I_2]}$$

where  $m$  is the modular ratio  $E_2/E_1$ .  $(I_1 + m I_2)$  can be treated as the equivalent second moment of area of the section. It can be used to produce an equivalent section by multiplying by  $m$  the dimensions of material 2 in the direction parallel to the neutral axis. The stresses are given by equation 2.

For the section shown in Figure 14 the compressive failing stress of a stringer plus effective skin is 263 MPa and the compressive failing stress of the corner members plus effective skin is 324 MPa as shown in Figure 15. The **effectiveness factor** for the stringer and skin is  $263/324 = 0.81$ . From Figure 15 it can be seen that this ratio can be considered to be the ratio of the modulus  $m$  of both items (assuming equal strains) at the failure load of the beam as  $\text{STRESS} = E \times \text{STRAIN}$ .

Therefore, treating the stringers in compression as material 2 in the above equations, then the area of stringers 2,3 and 4 plus skin must be multiplied by 0.81 to obtain their effective area as shown in table 2.

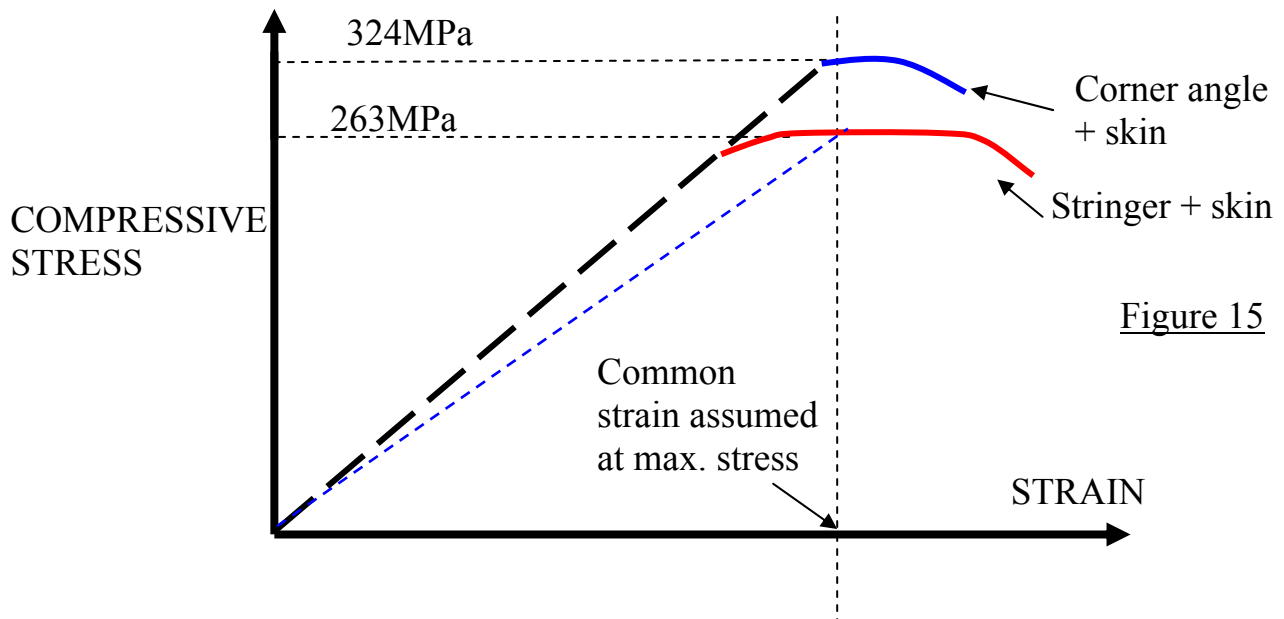
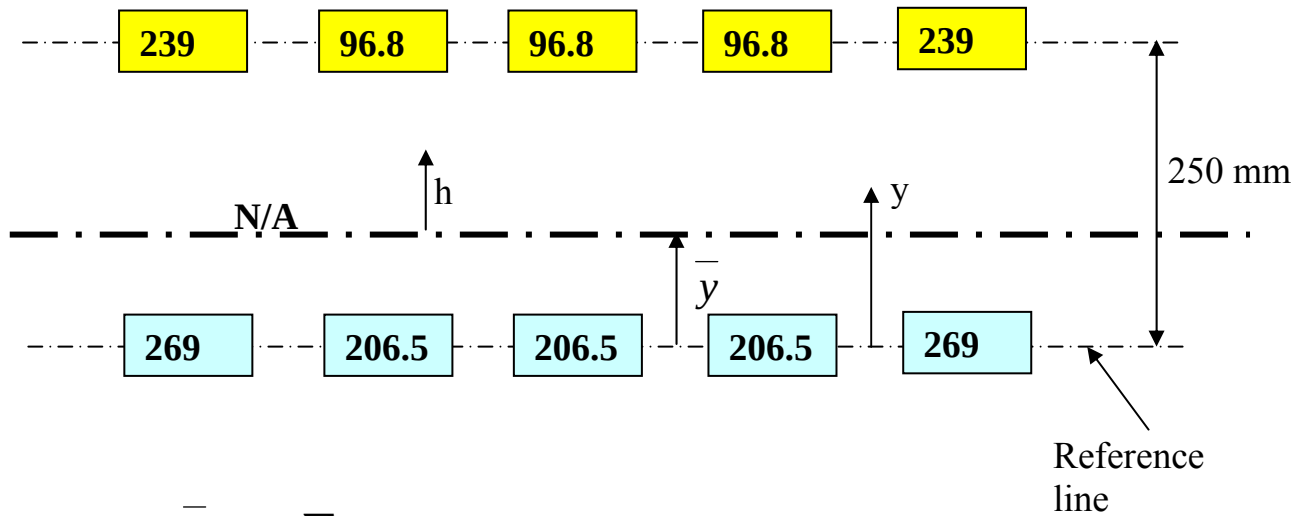


Figure 15

| Stringer<br>or<br>Angle | Stringer/angle area<br>plus effective skin<br>(mm <sup>2</sup> ) | Stringer/Angle<br>Effectiveness<br>Factor | Effective<br>Area<br>(mm <sup>2</sup> ) |
|-------------------------|------------------------------------------------------------------|-------------------------------------------|-----------------------------------------|
| 1                       | 239                                                              | 1.0                                       | 239                                     |
| 2                       | 119.5                                                            | 0.81                                      | 96.8                                    |
| 3                       | 119.5                                                            | 0.81                                      | 96.8                                    |
| 4                       | 119.5                                                            | 0.81                                      | 96.8                                    |
| 5                       | 239                                                              | 1.0                                       | 239                                     |
| 6                       | 269                                                              | 1.0                                       | 269                                     |
| 7                       | 206.5                                                            | 1.0                                       | 206.5                                   |
| 8                       | 206.5                                                            | 1.0                                       | 206.5                                   |
| 9                       | 206.5                                                            | 1.0                                       | 206.5                                   |
| 10                      | 269                                                              | 1.0                                       | 269                                     |

Table 2

Using the effective areas the location of the neutral axis and second moment of area of the equivalent section can be found.



$$\bar{y} = \frac{\sum Ay}{\sum A}$$

$$= \frac{(2 \times 239 + 3 \times 96.8) \times 250 + (2 \times 269 + 3 \times 206.5) \times 0}{(2 \times 239 + 3 \times 96.8 + 2 \times 269 + 3 \times 206.5)} = 99.74 \text{ mm}$$

Hence distance from neutral axis to centroid of edge members and stringers in compression =  $(250 - 99.74) = 150.26 \text{ mm}$ .

$$I_{NA} = \sum Ah^2 = [2 \times 269 + 3 \times 206.5] \times 99.74^2 + [2 \times 239 + 3 \times 96.8] \times 150.26^2 = 28.86 \text{ mm}^4$$

Ultimate bending moment =  $60.8 \times 10^6 \text{ Nmm}$

The compressive stress at the centroid of the corners/skin (material 1) is thus:-

$$\sigma_1 = \frac{My}{I} = \frac{60.8 \times 10^6 \times 150.26}{28.86 \times 10^6} = 316.5 \text{ MPa}$$

giving a reserve factor of  $324/316.5 = 1.02$ .

We can calculate the stress in the buckled stringers/skin using equation (2), namely:-

$$\frac{\sigma_1}{\sigma_2} = \frac{E_1 y_1}{E_2 y_2}$$

$$\sigma_2 = \frac{\sigma_1 E_2 y_2}{E_1 y_1} = \frac{316.5 \times 10^6 \times 0.81 \times 150.26}{150.26} = 256.4 \text{ MPa}$$

giving a reserve factor of  $263/256.4 = 1.02$ .

Note that both reserve factors are the same due to the calculated stresses being proportional to the “E” values which are themselves taken to be proportional to the failure stresses. However, the stringer/skin stress (256.4 MPa) is not the stringer/skin buckling stress (263 MPa) hence the effectiveness ratio will be higher than 0.81. More accurate stress values can be determined by iterating the ratio until the stringer/skin stress is calculated to be 263 MPa.

This simple example illustrates that the failure of real aircraft stiffened structures occurs under non-linear stress-strain conditions.

## Tutorial

**NOTE: In the following questions, where relevant, Figure 8 should be used to account for material plasticity effects.**

### Question 1

(a) Find, for the following two cases, the minimum value of the buckling coefficients (asymptotic values) and using these values calculate the buckling stress of a panel 100 mm wide.

- simply supported sides and ends
- fixed sides and ends.

Take  $E$  as  $63,000 \text{ N/mm}^2$  and a panel thickness of 0.9 mm.

(b) Determine, using Figure 8, the buckling stress of the panel for a thickness of 3 mm. for the simply supported case only.

### Question 2

(i) Calculate the Euler buckling stress for a strut, 1m long, cross section area of  $30\text{mm}^2$  and an  $I$  value of  $7180 \text{ mm}^4$ . Take  $E$  as  $67,692 \text{ N/mm}^2$  as shown for the material in Figure 7. Assume pinned ends.

(ii) Determine the tangent modulus at the Euler buckling load using Figure 7 and hence the new Euler buckling stress.

### Question 3

For the panel section shown in Figure Q3 determine:-

- The critical stress values for skin, flange, web and flexural instability.
- the effective skin width
- the total load capacity of the section

K coefficients:-

Skin: 4.1, web: 3.62, free flange: 0.58

Rib Pitch : 700 mm

For stringer plus effective width (30 t) the radius of gyration = 15.2 mm

Assume that the ends of panel act as though they are pinned.

$$\frac{d\sigma_{AV}}{d\sigma_E} = 0.333$$

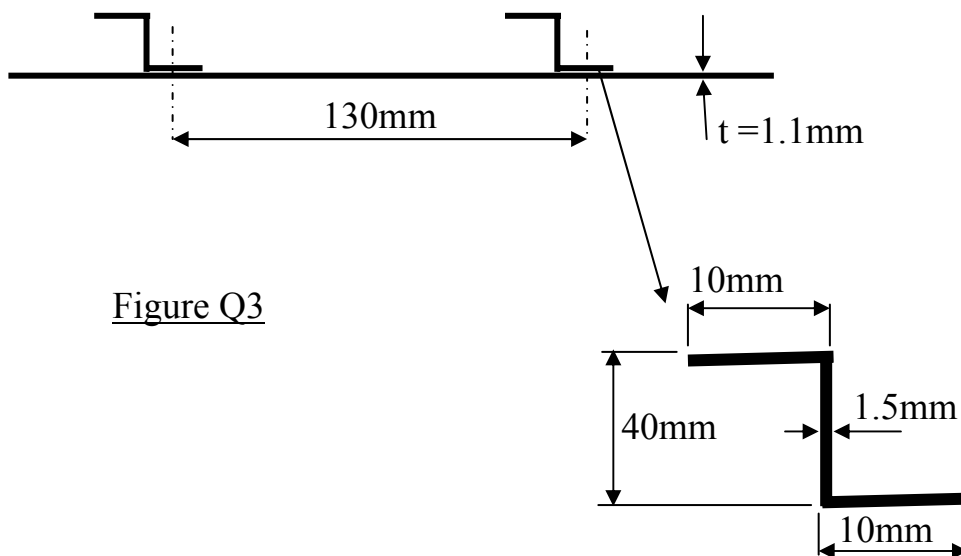


Figure Q3

### Question 4

A double skinned compression panel is shown in Figure Q4. One skin is 1mm thick and the other skin is 0.8mm thick. The two skins are riveted to “Z” stiffeners set at a pitch of 100mm. The skin stiffener rivet pitch is 15mm. The length of the panel is 0.6 m and can be considered to be pin jointed at each end.

Buckling coefficients of the panel components are:-

- skin: 4.2
- stiffener web: 3.62
- stiffener flange: 0.58
- inter-rivet: 3

$\frac{d\sigma_{AV}}{d\sigma_E} = 0.3$  and the radius of gyration of a stringer with its effective skin is 25mm.

Calculate :-

- the maximum compressive load that can be carried by a skin/stiffener section before skin buckling occurs.
- the compressive failure load of a skin/stiffener section.

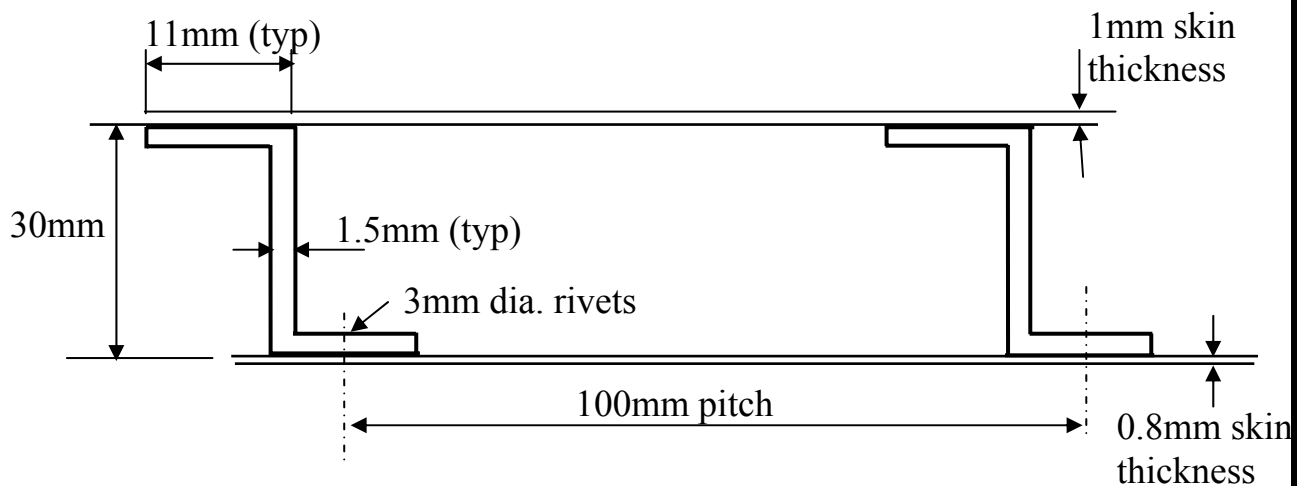


Figure Q4

### Question 5

The wing compression panel shown in Figure Q5 comprises a 0.8mm thick sheet stiffened by 'I' sectioned stringers. The stringers can be considered to be simply supported at a pitch of 400mm by the wing ribs. The radius of gyration (k) for one stringer section plus a 30x skin thickness length of skin is 16mm.

The skin buckling stress coefficient is 4.2, the stringer web buckling stress coefficient is 3.62 and the coefficient of the free flanges is 0.58.

The ratio  $\frac{d\sigma_{AV}}{d\sigma_e} = 0.32$

Calculate for one skin stringer section:-

- (i) the skin buckling stress
- (ii) the buckling stress of the stringers
- (iii) the failure load of the panel
- (iv) the buckling stress of the web required to give a panel failure load of 26,500N.

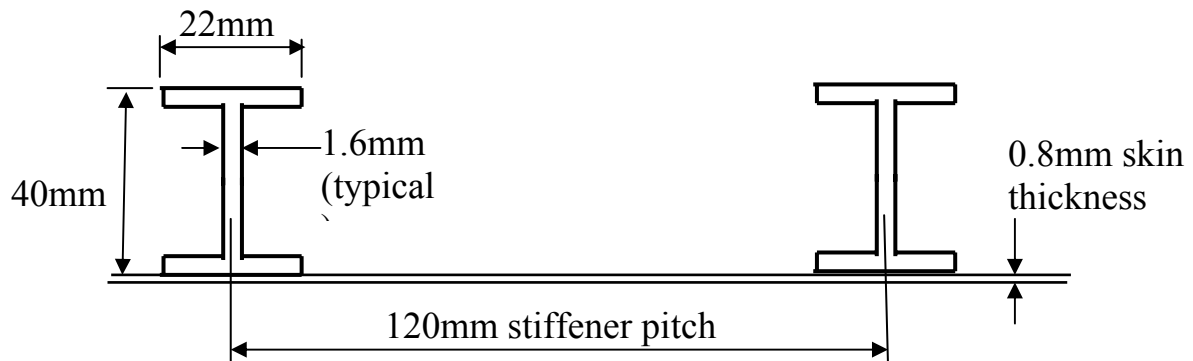


Figure Q5

ANS: (i) 11.8N/mm<sup>2</sup>  
(ii) 152  
(iii) 25.1  
(iv) 160.7

### Question 6

Determine the reserve factor in the compression members of the wing box section shown in Figure Q6 at the ultimate bending load of 33.5 kNm.

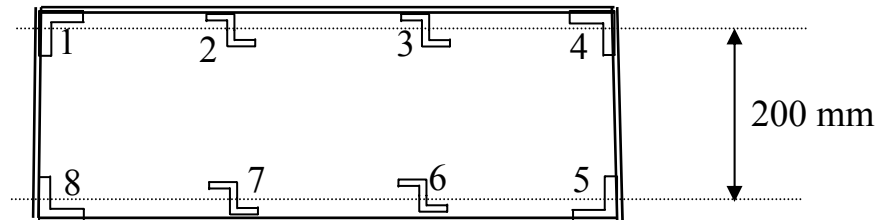


Figure Q6

Area of stringers 2 and 3 plus effective skin =  $100 \text{ mm}^2$  (each) and buckling stress is 250 MPa.

Area of corner angles 1 and 4 plus effective skin =  $200 \text{ mm}^2$  and buckling stress is 300 MPa.

Area of corner angles/skin in tension =  $230 \text{ mm}^2$

Area of stringers/skin in tension =  $150 \text{ mm}^2$

Assume that all area centroids lie on the 200 mm separated lines shown in Figure Q6